

INTERVIEW QUESTIONS & RESPONSES

Computer Networks - CSA07

Student Name: S Hariram

Register Number: 1965260018

Q1. Question Asked:

What is the difference between the roles of SMTP and POP3 when sending and receiving electronic mail?

Student Response:

SMTP is a push protocol used to send emails from a client to a mail server, or between mail servers. It operates on port 25, 465, or 587. POP3, on the other hand, is a pull protocol used by the recipient's mail client to retrieve emails from the mail server, downloading them locally. It typically operates on port 110 or 995.

Q2. Question Asked:

A user can access a website using its IP address but not its domain name. Which application layer protocol is likely failing, and how would you troubleshoot it?

Student Response:

The failing protocol is DNS (Domain Name System), which resolves domain names to IP addresses. To troubleshoot, I would use tools like `nslookup` or `dig` to verify if the DNS server is responding correctly. I'd also check the client's network configuration for the correct DNS server IP, clear the local DNS cache, and ensure no firewall is blocking port 53.

Q3. Question Asked:

How does a Peer-to-Peer (P2P) network differ from a traditional Client-Server architecture, particularly regarding scalability?

Student Response:

In a Client-Server architecture, clients request resources from a central server, which can become a bottleneck under heavy load. In a P2P network, each node acts as both a client and a server (a peer). This makes P2P highly scalable because as more peers join the network, the overall bandwidth and resource capacity organically increase, distributing the load.

Q4. Question Asked:

VoIP calls often experience jitter. What is jitter, and how can overlay networks or specific protocols help mitigate its effect on call quality?

Student Response:

Jitter is the variation in packet arrival times, which causes choppy audio. VoIP systems mitigate this using jitter buffers at the receiving end to collect and sequence packets before playing them at a steady rate. Protocols like RTP (Real-Time Transport Protocol) provide timestamps and sequence numbers, while overlay QoS mechanisms prioritize voice traffic to ensure consistent delivery.

Q5. Question Asked:

Explain the role of the SSL/TLS handshake in establishing a secure communication channel. What key cryptographic concepts are involved?

Student Response:

The handshake establishes an encrypted connection. The client and server agree on a TLS version and cipher suite. The server sends its digital certificate containing a public key for authentication. Asymmetric encryption is used to securely exchange a shared master secret, which is then used to generate symmetric keys for efficiently encrypting the actual data stream, ensuring confidentiality and integrity.

Q6. Question Asked:

How does a stateful firewall differ from a stateless firewall, and which is better equipped to mitigate a TCP SYN flood DoS attack?

Student Response:

A stateless firewall inspects individual packets based on static rules (like IP/port) without context. A stateful firewall maintains a state table of active connections and understands the context, like the TCP 3-way handshake. A stateful firewall is much better equipped to mitigate a SYN flood because it tracks half-open connections and can enforce limits on incomplete handshakes.

Q7. Question Asked:

You are tasked with securing an e-commerce website that currently uses HTTP. Walk me through the necessary protocols and mechanisms you would implement to ensure customer data is protected.

Student Response:

I would immediately migrate from HTTP to HTTPS by installing an SSL/TLS certificate from a trusted CA. I'd implement HTTP Strict Transport Security (HSTS) to force all connections over HTTPS. Additionally, I'd configure a Web Application Firewall (WAF) to protect against common web attacks and ensure session cookies use the 'Secure' and 'HttpOnly' flags.

Q8. Question Asked:

An organization's web server is crashing due to a massive influx of DNS response packets, even though the server hasn't sent out DNS queries. What type of attack is this, and how can it be mitigated?

Student Response:

This is a DNS Amplification Attack, a type of reflection DDoS. The attacker spoofs the victim's IP and sends small queries to open DNS resolvers, which reply with large responses to the victim. To mitigate this, ISPs should implement ingress filtering (BCP38) to drop spoofed traffic, and the organization can rate-limit inbound UDP port 53 traffic or use a DDoS protection service.

Q9. Question Asked:

Email spoofing is a common method for phishing attacks. What mechanisms exist to prevent attackers from sending emails that appear to come from your organization's domain?

Student Response:

To prevent email spoofing, organizations use SPF, DKIM, and DMARC. SPF specifies which IPs can send mail on behalf of the domain. DKIM adds a cryptographic signature to verify the sender and message integrity. DMARC ties them together, instructing receiving servers on what to do (like reject or quarantine) if an email fails the SPF or DKIM checks.

Q10. Question Asked:

How could a secure overlay network, such as a VPN, impact the performance of a real-time VoIP application, and how would you optimize the network to balance security and performance?

Student Response:

A VPN adds encryption and encapsulation overhead, which can increase latency and jitter, hurting real-time VoIP. To optimize, I would use a lightweight, UDP-based VPN protocol like WireGuard or OpenVPN over UDP to avoid TCP retransmission delays. Implementing QoS to prioritize VoIP packets before they enter the tunnel also helps maintain call quality while staying secure.